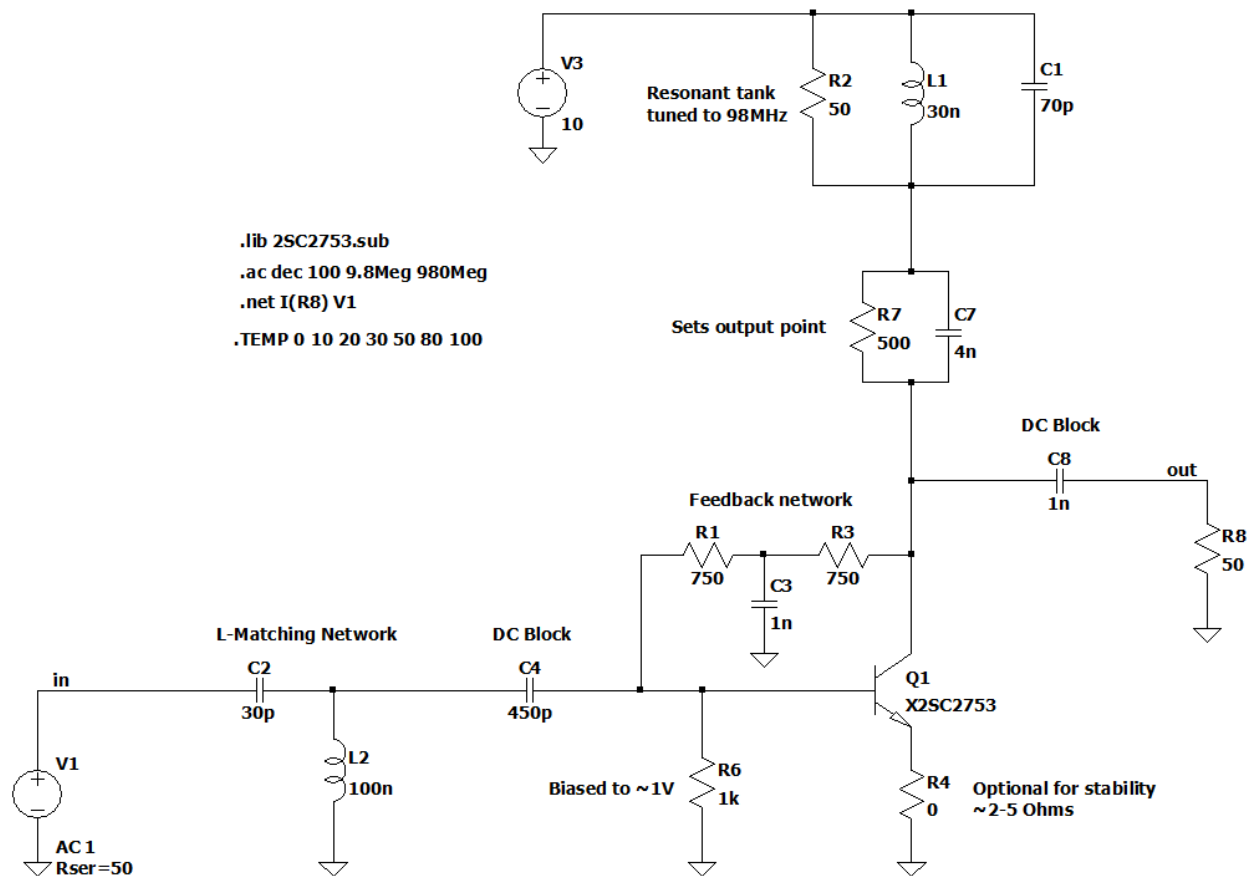


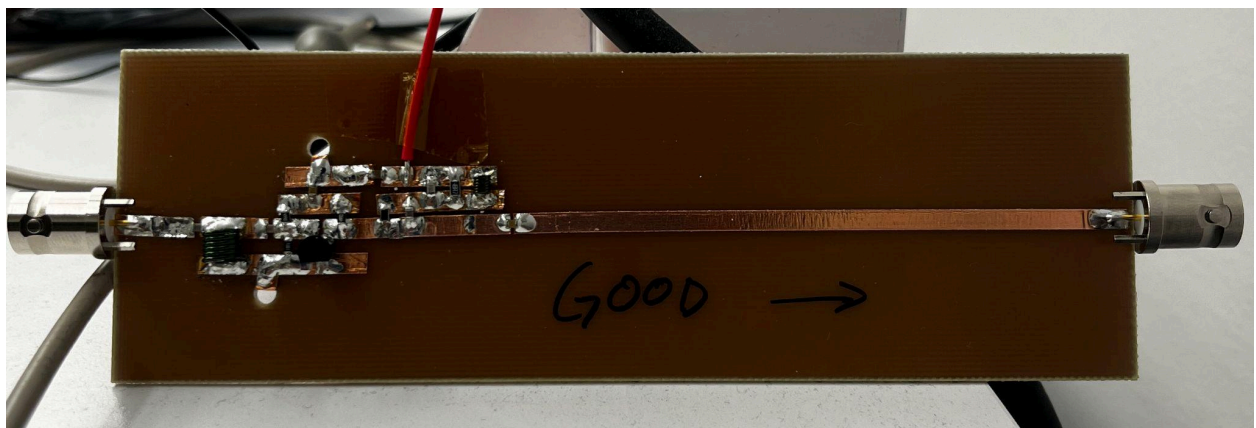
Date: 02.04.2026

	88MHz	98MHz	108MHz
S11	-15.6dB	-21.7dB	-22.3dB
S21	19.3dB	18.9dB	18.1dB
S22	-13.7dB	-15.7dB	-14.0dB

Schematic diagram of narrow band amplifier.



Photograph of implemented narrow band amplifier.



Technical Discussion:

A common emitter amplifier topology used. There is no emitter degeneration since we want high gain. However for temperature stability a small emitter resistor can be used.

We want ~10mA through the transistor and with 10V supply the output should be at 5V so it has equal swing in either direction. This sets the collector resistor as ~500ohms

$$R_C = V_{R_C} / I_C = (10V - 5V) / 10mA = 500\Omega$$

Next the RLC tank is tuned to resonate at 98MHz. Arbitrarily choose 30nH for an inductor value and calculate the capacitor value.

$$\omega = 1/\sqrt{LC} \rightarrow C = 1/(\omega^2 L) = 1/((2\pi \cdot 98MHz)^2 \cdot 30nH) = 87pF$$

In practice the LRC tank circuit needed to be tuned to meet spec. This was done by adjusting the capacitance and inductor values to change the resonant frequency present on the board. A 50 ohm resistor in parallel is used since it produced good peaking that wasn't narrow enough to exclude frequencies of interest.

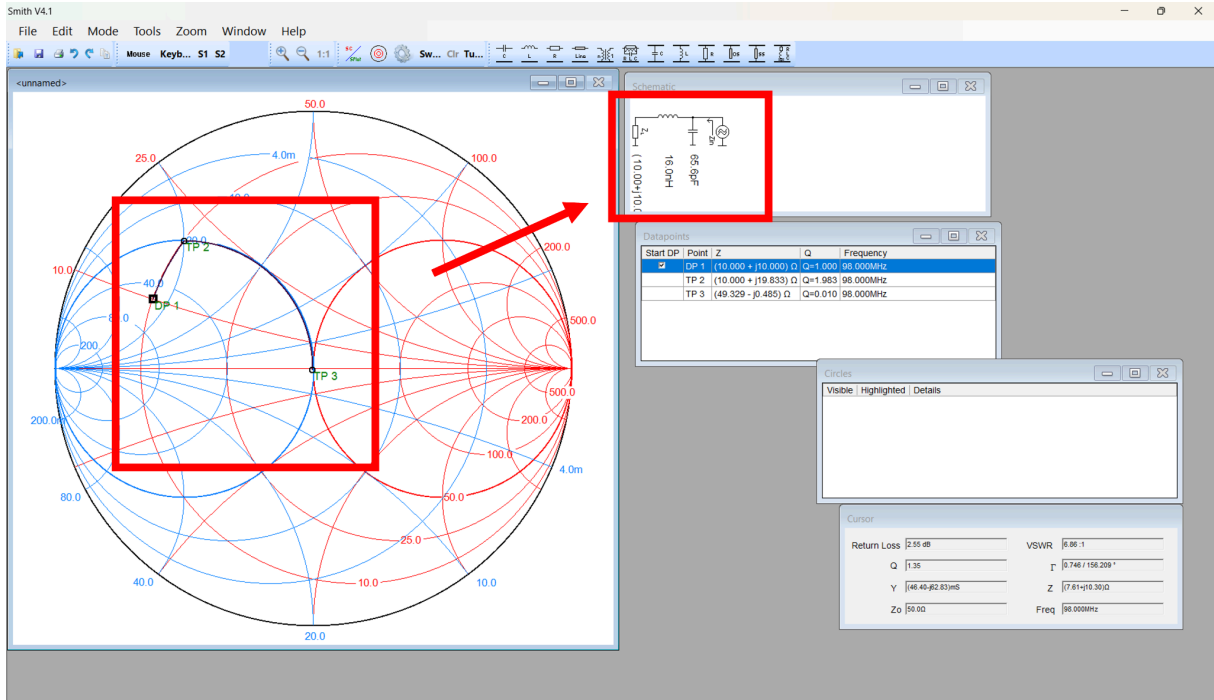
The bias point of the transistor is set at 1V. A feedback network connecting the base to the collector through a resistor network also helps set the bias point. The current flowing through the feedback network should be roughly 10 times smaller than the collector current. Using this spec the current through the base bias resistor is roughly 1mA. Since we want a 1V bias point the bias resistor is 1k.

The output of the amplifier is set at 5V so there is a 4V drop across the feedback network. This requires a few thousand ohms of resistance.

$$R_f = (5V - 1V) / 1mA = 4V / 1mA = 4k\Omega$$

In practice we found that this feedback resistance is too high and had to reduce to a 1.5k feedback resistance. The feedback network is also tapped to help prevent feed-forward of the desired signals.

An L-match is used at the input to help impedance matching. The network was derived using a smith chart and the complex impedance reported on the VNA at the input port. Smith4.1 is a software that lets you enter complex impedances and add devices to achieve a desired match. The image below shows the software and the resulting network for the inputted complex impedance.



An additional DC blocking cap is placed between the input L-match and the rest of the amplifier. This value was initially 1nF but was adjusted down to roughly half of that value (450pF) which gave the amplifier better performance.

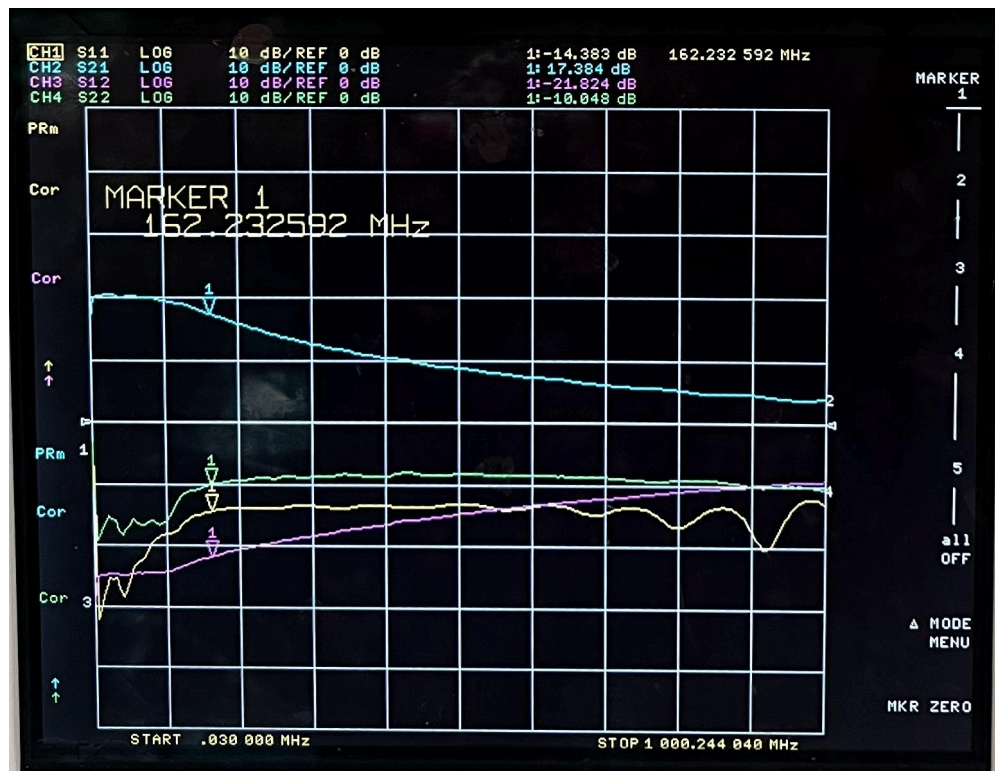
Broadband Amplifier

We set out to a broadband amplifier with as wide of bandwidth as possible while providing 15db of gain with a 50 ohm impedance matched input and output.

Performance:

S11	S21	S22
> 15db from 30KHz to 220Mhz	< 10db	< 10db for less than 162MHz

VNA Output:



LTSpice Schematic:

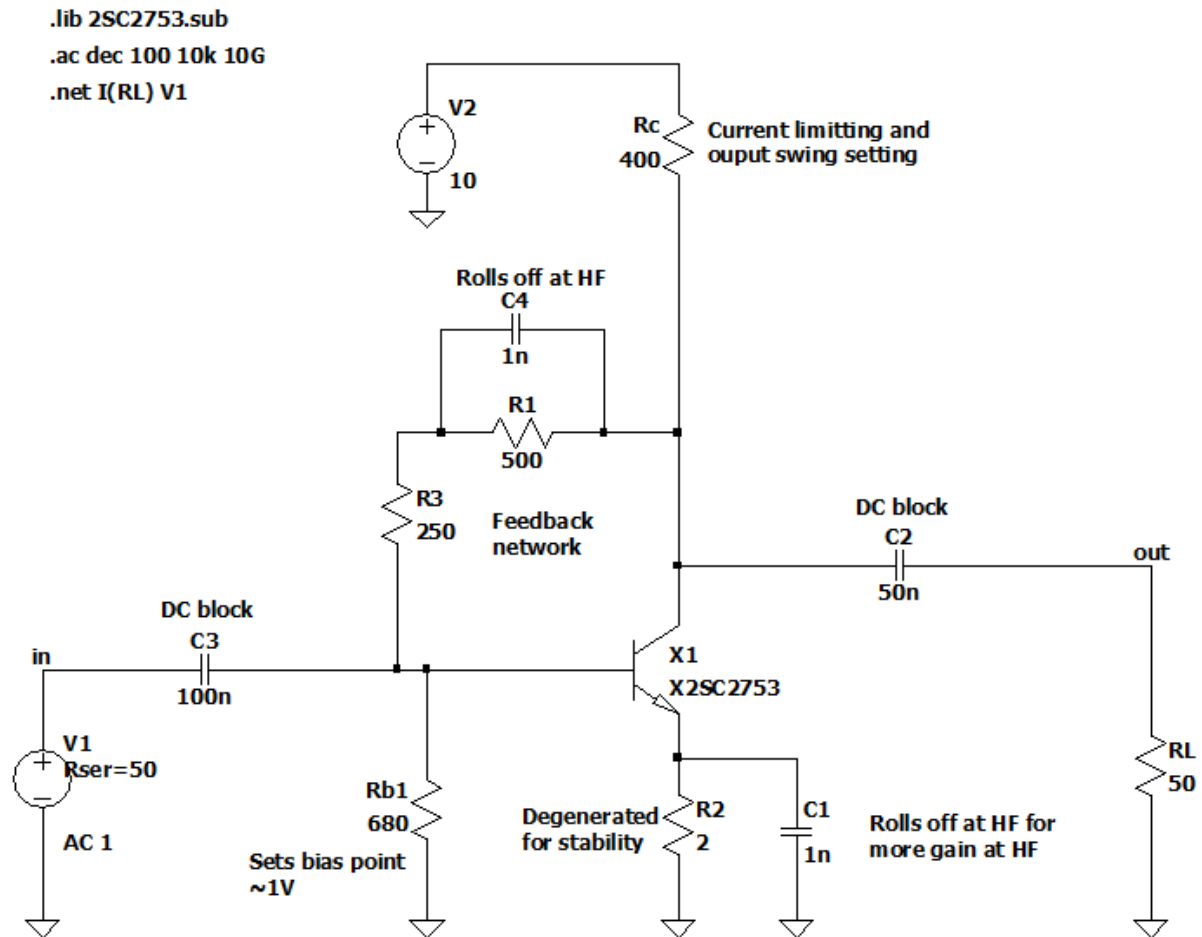
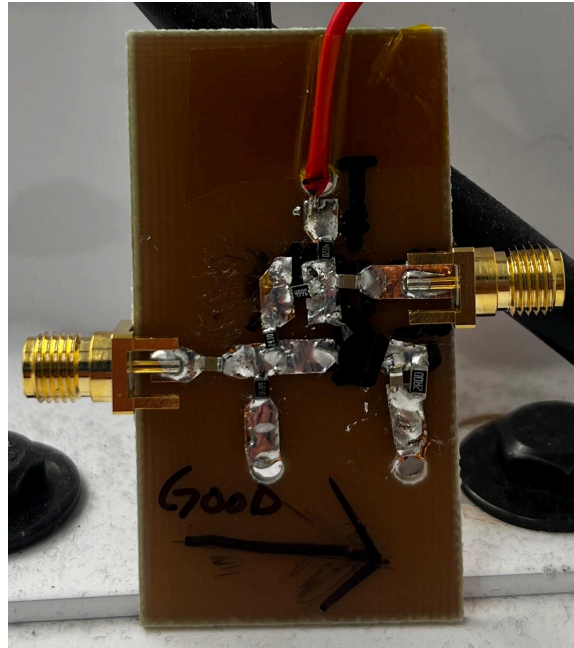


Photo of Broadband Amplifier:



Technical Discussion:

The broadband architecture is a Shunt-Series Feedback Amplifier, which is a standard configuration for wideband applications. In the Shunt Feedback ($R1 + R3$, $C4$), the feedback resistor $R1 + R3$ samples the output voltage and returns a current to the base. In addition, in the Series Feedback ($R2$, $C1$), the emitter resistor $R2$ samples the output current and provides emitter degeneration which increases linearity.

In order to get a wide bandwidth, one of the tricks from the textbook we used was pole-zero cancellation where a small capacitor, $C4$ in our schematic, is placed in parallel with the feedback resistor $R1$. As the frequency increases, the impedance of the feedback path decreases, allowing the amplifier to maintain a flatter and higher gain at higher frequencies. 750 ohms to 250 ohms effective resistance at higher frequencies. Also we added $C1$ across the emitter resistor $R2$ to reduce the degeneration at higher frequencies, helping to boost the gain.

One calculation we made was for the feedback resistor R_F in figure 12.28 in the textbook. We are looking for greater than 15db gain, so a gain magnitude upwards of 5.62. Using equation (52), we calculate $R_F \approx Z_0(1 - AV) = (50)(1 + 5.6) = 330$. However we traded this with a 250 ohm resistor for more bandwidth.

Also, to avoid excessive overloading we want our R_{B1} to be more than 10 times the source resistance, meaning at least 500 ohms, but less than 1k ohms in order to allow at least 10 times the base current. We used 680 ohms as used in the book because it lies roughly in the middle. In addition, when selecting our collector resistor to drop half the supply voltage, we opted to reduce R_L from 500 ohms to 400 ohms in order to target a higher bias current of approximately 20mA and to achieve a greater gain bandwidth product. Our LTSpice model operating point

calculates our bias current at 18.3mA, giving us a 7.3 volt drop across R_L , thus leaving our $V(\text{collector-emitter})$ to $10 - 7.32 - 0.037 = 2.64$ volts, ensuring linear mode operation.

Regarding matching, the input impedance is matched using shunt-series resistive feedback rather than LC matching networks. The feedback resistors from collector to base, and with emitter degeneration, help transform the input impedance toward 50 ohms over a broad frequency range. At some point, we did try implementing an L-match at the output to improve our S_{22} parameter, however this increased peaking and reduced our amplifier bandwidth, without much S_{22} performance gain.

For calculating the specific OCTC for our feedback network configuration, it includes a 250 ohm resistor R_F in series with a 500 ohm resistor R_{B2} , that also has a 1nF capacitor C_B in parallel. Here we treat the capacitor C_B as an open circuit to perform the calculation. Therefore, our effective feedback resistance is 750 ohms. Next our V_{CC} is 10V and our R_C is 400 ohms. This gives us a bias current of 23mA, and using V_T from the datasheet our transconductance is 0.88S. Our transistor input resistance is $100/0.88$ which is 133 ohms. Using the datasheet our collector-base parasitic capacitance is 0.65pF and our base-emitter parasitic capacitance derived from $g_m/2\pi f_T - C(\text{collector-base})$ is 27.4pF. Next, our resistance looking into the base-emitter junction is just our 50 ohm source resistance in parallel with the transistor input resistance which is 34.6 ohms. Then our resistance looking in to the collector-base junction because of the Miller Effect is $R(\text{Miller}) = R_{in}(1 + G_m R_{out}) + R_{out} = 34.6 (1 + (0.32 * 44.4)) + 44.4 = 570$ ohms which is in parallel with our effective feedback resistance 750, which equals 324 ohms. To estimate our bandwidth, we find the base-emitter time constant, which equals $R_{in} * \text{base-emitter capacitance} = 34.6 \text{ ohms} * 27.4 \text{ pF} = 948$ ps. Then our collector-base time constant is our feedback resistance * collector-base parasitic capacitance which equals $324 \text{ ohms} * 0.65 \text{ pF} = 210$ ps. Finally, we add the two time constants and multiply by 2π to convert to frequency and then divide this product under 1. This gives us an estimated -3db bandwidth of 137MHz.

This estimated bandwidth is fairly close to our measured 162 MHz bandwidth. However, the discrepancy in our higher measured bandwidth comes from the fact that the OCTC calculations assume the feedback resistance is a constant 750 ohms. But our addition of the 1nF C_B capacitor in parallel with the 500 ohm resistor shorts out that resistor at higher RF frequencies, reducing the feedback impedance to 250 ohms, which extends the bandwidth beyond the estimated 137MHz.